6 **(a)** Explain what is meant by *binding energy* of a nucleus.

.....

.....

..... [1]

(b) **(i)** Give a sketch on Fig. 6.1 to show the variation of the binding energy per nucleon with nucleon number.

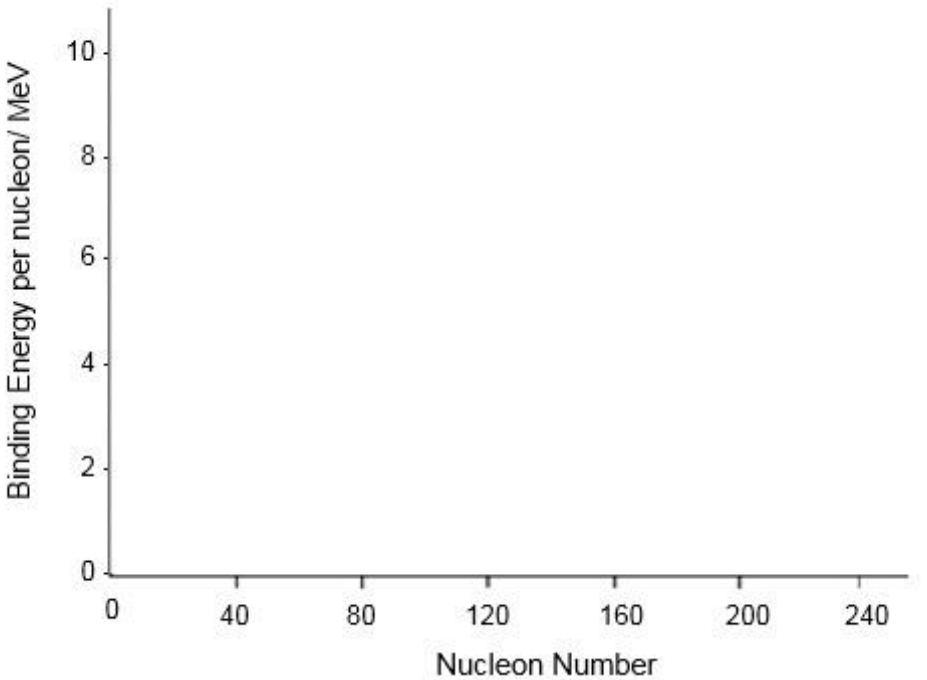

Fig. 6.1 [2]

(ii) With reference to your sketch in Fig. 6.1, explain how fission can be a potential source of energy.

.....

.....

..... [1]

(c) When a Uranium-235 nucleus undergoes fission, two nuclides are produced with the release of energy as shown in the equation below.

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{139}_{54}Xe + ^{95}_{38}Sr + 2\,^{1}_{0}n$$

The masses of the nuclides are as follows:

nuclide	mass
$^{235}_{92}U$	235.043929 u
$^{139}_{54}Xe$	138.918793 u
$^{95}_{38}Sr$	94.919359 u
$^{1}_{0}n$	1.008665 u

(i) Explain how a chain reaction is able to occur for this nuclear fission.

.....

..... [1]

(ii) Calculate the energy released in one reaction.

energy released = ... J [3]

(iii) Singapore's energy consumption in 2017 was approximately 50 TWh. Assuming an efficiency of 8.0 %, determine how long (in months) that the energy released from the fission of 2.0 kg of Uranium can be used to power Singapore.

time = ... months [4]